Water Wheel

Flagship Lab 2 · Grade 4 Using Natural Resources for Energy · Student Sheet

Name: _____ **Date:** _____

BIG QUESTION **How does falling water transfer energy to lift a load?**

WHAT WE KNOW

Falling water has energy of motion. That energy can be transferred to a wheel and used to lift a load.

FAIR TEST

KEY IDEA **We change ONE thing. We keep everything else the same.**

THE ONE THING WE CHANGE	THE ONE THING WE MEASURE
pour height (how high the water is poured from)	how many paperclips the wheel lifts

KEEP THE SAME

These parts must stay the same in every trial:

☐ wheel design ☐ axle ☐ blade setup

☐ water volume ☐ pour opening ☐ thread / load system

PREDICTION

If I pour water from a HIGHER height, I predict the wheel will lift _____ paperclips.

DRAW THE WHEEL

Draw your group's wheel. Label the blades, the axle, and the thread.

draw your water wheel here

Test and Record

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DATA TABLE

Test three pour heights. Keep everything else the same. Record what happens.

TRIAL	POUR HEIGHT (CM)	PAPERCLIPS LIFTED (COUNT)	WHAT I NOTICED
Low			
Medium			
High			

ENERGY PATHWAY

Number the pathway in order from 1 to 4.

STEP	ORDER
falling water	
spinning wheel	
water at the top	
lifted load	

Claim and Redesign

Name: _____ **Date:** _____

MAKE A CLAIM

Finish the claim using your data.

When I poured water from a higher height, the wheel lifted _____ paperclips.

This is evidence that falling water from a higher height has more energy of motion to transfer.

REDESIGN

If you could change ONE part of the wheel to lift more paperclips, what would you change? Why?

CHECK YOUR THINKING

A real hydroelectric dam uses spinning turbines connected to generators. How is your water wheel like a dam's turbine?

CONNECT *Water wheel connection: falling water → spinning wheel → lifted load. Real dams: falling water → spinning turbine → generator → electricity.*